Electrical Power and Energy Systems 164 (2025) 110335

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Distributed secondary control for DC microgrids using two-stage multi-agent reinforcement learning

Fei Li [a], Weifei Tu [a], Yun Zhou [b,*], Heng Li [c,*], Feng Zhou [d], Weirong Liu [c], Chao Hu [c]

[a] School of Automation, Central South University, Changsha, China
[b] Hunan University of Finance and Economics, Changsha, China
[c] School of Electronic Information, Central South University, Changsha, China
[d] School of Electrical and Information Engineering, Changsha University of Science and Technology, Changsha, China

ARTICLE INFO

Keywords:
DC microgrids
Multi-agent system
Deep reinforcement learning
Secondary control

ABSTRACT

Multi-agent reinforcement learning has emerged as a promising candidate for the secondary control of DC microgrids. However, the one-stage reward function incorporating both voltage regulation and current sharing results in the significant bus voltage fluctuations and long current sharing time. To address this issue, in this paper, we propose a two-stage reinforcement learning secondary control method for DC microgrids, which can effectively suppress the bus voltage fluctuations and reduce the current sharing time. The multi-agent Proximal Policy Optimization (PPO) algorithm is utilized to regulate the current and voltage of each node in the microgrids. Specifically, a two-stage reward function based on voltage error and current error is designed, which can effectively improve the convergence speed. Moreover, an action safe mechanism is constructed to mitigate the effects of random noise and ensure the smooth operation of the DC microgrids. We have built a hardware-in-the-loop platform to verify the effectiveness of the proposed method. Experiment results show that the proposed method can effectively improve the current sharing speed and reduce the bus voltage fluctuation when compared with existing methods.

1. Introduction

Due to the lack of traditional fuels and the booming development of renewable energy sources(RESs), microgrids have been in the focus as one of the ways to integrate renewable energy generation in recent years [1,2]. According to the bus voltage type, microgrids are mainly divided into two categories: AC microgrids and DC microgrids [3]. And DC microgrids have received great attention from researchers due to their straightforward design, simple operation, and high energy efficiency [4,5]. DC microgrids are usually equipped with energy storage systems (ESSs) for RESs to effectively reduce microgrid swings caused by random fluctuations of RESs, so a stable and efficient control strategy for DC microgrids is especially important [6,7].

Generally, the control strategy of DC microgrids mainly consists of two layers of control: the primary control and the secondary control. The primary control always used droop control as a current sharing method [8]. It should be noted that the traditional droop control cannot achieve current sharing and voltage regulation simultaneously due to line impedance mismatch, sensor sample error, etc. For this reason, the secondary controller was developed to focus on the energy coordination between DC microgrids [9,10].

Currently, in secondary control of DC microgrids, the control strategies are generally divided into centralized control and distributed control [11]. Centralized control is able to ensure steady bus voltage. However, centralized control necessitates the use of a central master controller to gather and process the state information of each distributed generation, which can cause a serious communication burden and sometimes suffer from a single point of failure, and may be hindered by dimensionality of system states [12].

As a result, distributed control strategies have been developed [13]. It has multiple local controllers, optimizes the respective control strategy by sharing information with its neighbors and improves the robustness of the control system [14]. Mokhtar et al. proposed updated and harmony search-optimized droop resistances to minimize current sharing errors and communication needs in DC microgrids for load current sharing, demonstrating effective performance in simulations [15]. Zeng et al. proposed a current regulator to provide a current sharing compensation term and a voltage observer to provide a voltage compensation term as a distributed secondary control [16]. Zhang et al. designed an information state factor containing voltage and current to achieve secondary control of DC microgrids and reduce the communication

* Corresponding authors.
E-mail addresses: zhouyun0110@vip.163.com (Y. Zhou), liheng@csu.edu.cn (H. Li).

https://doi.org/10.1016/j.ijepes.2024.110335
Received 25 February 2024; Received in revised form 27 August 2024; Accepted 22 October 2024
Available online 19 November 2024

Table 1
Comparison of secondary control strategies of DC microgrids.

Methods	Voltage recovery	Current sharing speed	Current sharing accuracy	Required parameters adjustment	Safety and stability	Communication reliance
Droop	✗	Fast	Low	Few	High	None
Centralized consensus control	✓	Slow	High	Many	High	High
Distributed consensus control	✓	Slow	High	Many	High	Weak
MADDPG	✓	Fast	High	Few	Low	Weak
Proposed	✓	Fast	High	Few	High	Weak

burden [17]. In these studies, PI controllers are used to improve the accuracy of current sharing and voltage regulation. However, all the above-mentioned studies are based on PI controllers with delay effect of the integrating link and are sensitive to system parameters such as line impedance, controller parameters, and virtual impedance.

In recent years, reinforcement learning (RL) has been widely used in DC microgrids with its better adaptive ability and dynamic performance [18]. Barbalho et al. used deep deterministic policy gradient (DDPG) to change the output power of the storage elements to secure the microgrid voltage and frequency stability [19]. In addition, a dual-time ratio voltage controller was proposed in [20], where parallel capacitors were configured to minimize voltage error using a deep reinforcement learning algorithm. A control algorithm utilizing Twin Delayed Deep Deterministic Policy Gradient and DDPG was proposed in [21] to support frequency in low-inertia grids, showing improvements over conventional methods in terms of RoCoF and nadir. However, the above literatures are all centralized voltage control frameworks using a single agent and still have the aforementioned shortcomings of the centralized control strategy.

Multi-agent Reinforcement Learning (MARL) has made significant progress in recent years due to its ability of traditional reinforcement learning in multi-agent environments' Non-stationarity and partial observability [22,23]. MARL has been widely used as a distributed method in various complex systems. Wu et al. proposed a multi-timescale voltage control scheme based on MADDPG with discrete action output control signals [24]. Sun et al. proposed a two-stage Volt-Var control, and after the energy scheduling in the first stage, the real-time Volt-Var control in the second stage was implemented based on MADDPG [25]. Xia et al. optimized the traditional MADDPG to deal with the secondary control of islanded AC microgrids and used a global control objective function instead of the critic network to improve the control performance of the algorithm [26]. However, in these studies, the one-stage reward function that includes both voltage regulation and current sharing is difficult to explore the optimal control strategy, which is prone to cause significant fluctuations in bus voltage and long current sharing time.

Table 1 shows a comprehensive comparison study highlighting the differences between the control strategy proposed in this paper and the existing secondary control strategies in DC microgrids. Motivated by the above, in this paper, we propose a two-stage multi-agent reinforcement learning method for the secondary control of DC microgrids. In the proposed method, the voltage regulation term and current sharing term are prioritized in two stages to suppress the voltage fluctuation and improve the current sharing speed. Different from existing studies, the proposed method utilizes the Proximal Policy Optimization (PPO) algorithm [27]. The PPO algorithm can provide continuous control output and thus has wide applications [28,29]. We have built a hardware-in-the-loop platform to verify the effectiveness of the proposed method. Experiment results show that the proposed method can effectively improve the current sharing speed and reduce the bus voltage fluctuation when compared with existing methods.

The main contributions of this paper are summarized as follows:

- A two-stage multi-agent reinforcement learning method is proposed for DC micro-grids, which can effectively smooth the bus voltage fluctuations and improve the current sharing speed.
- The proposed method does not require droop control in the primary control layer and does not need accurate model parameters, and numerical results have validated its robustness and stability.
- We have built a hardware-in-the-loop platform, and extensive experiment results show that the proposed method can effectively improve the current sharing speed and reduce the bus voltage fluctuation when compared with existing methods.

The remainder of the paper is organized as follows. The preliminaries and problem statement are presented in Section 2. In Section 3, the proposed secondary control is introduced. Experimental results and discussions are presented in Section 4 whereas concluding remarks and future works are drawn in Section 5.

2. DC microgrids system environment and problem statement

In this section, a DC microgrid environment with multiple sub-microgrids is introduced to help us better establish a basic experiment environment including PV and storage battery, which is used to verify the effectiveness of the control strategy proposed in this paper and to compare with other control strategies. Subsequently, the information flow is presented within the communication network based on undirected connected graph theory and the power flow is based on the traditional Consensus control method. Lastly, the secondary control problem is discussed in this section.

2.1. DC microgrids system environment

As shown in Fig. 1, the DC microgrids studied in this paper consist of multiple DC sub-microgrids and each one contains the load, the Photovoltaic (PV) generation modules and the storage battery. Additionally, the line impedance is considered between each sub-microgrid in the system environment. To fully utilize the renewable energy generation capacity, the PV generation modules are operated in Maximum Power Point Tracking (MPPT) mode. And when the available power exceeds the load demand, the excess power is used to charge the storage battery. In contrast, the storage battery will fill in the gap when the load demand exceeds the generated power. Thus, the storage battery plays a vital role in peak shaving and valley filling, ensuring the stable operation of the system.

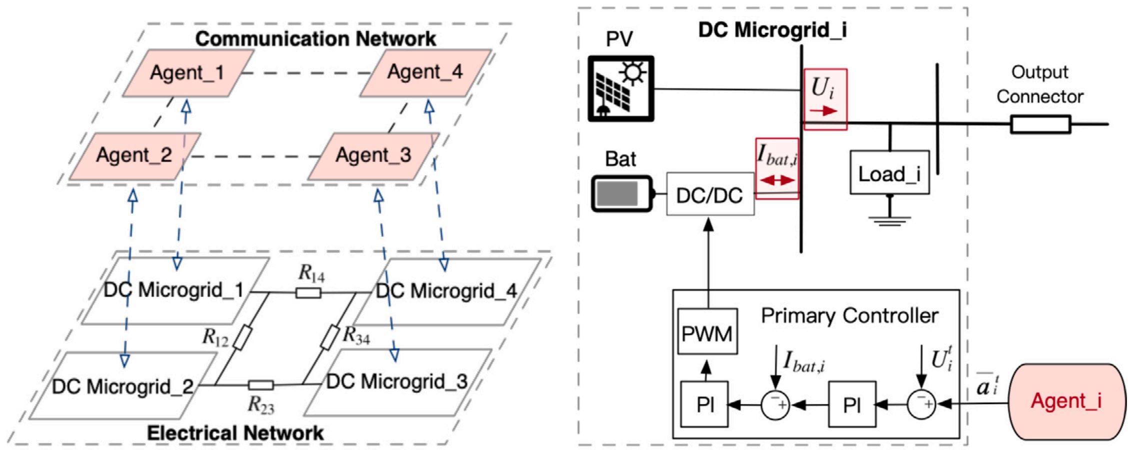
In this paper, photovoltaic is used as a renewable energy generation device in the studied DC microgrid, and the device model can be referred to as [30]. The strategy proposed in this paper is for the secondary control strategy of the storage battery, which is also suitable for other renewable energy generation devices.

In addition, the electrical circuit model of the storage battery considered in this paper is composed of an equivalent internal resistance R_{sb} and a controlled voltage source E_{bat}, as shown below:

$$E_{bat} = E_0 - k\frac{Q}{Q - \int_0^t I_{bat}dt} + Aexp\left(-B\int_0^t I_{bat}dt\right) \tag{1}$$

where E_0 is the constant voltage; k is the polarization factor; Q is the storage battery capacity; I_{bat} is the storage battery current; A is the amplitude corresponding to the exponential discharge zone; B is the reciprocal of the time constant of the exponential discharge zone. And the storage battery voltage U_{bat} is:

$$U_{bat} = E_{bat} - R_{sb}I_{bat} \tag{2}$$



(a) The structure of secondary control of DC microgrids (b) The structure of secondary control of sub-microgrid

Fig. 1. The structure of secondary control of DC microgrids and each sub-microgrid.

2.2. Information flow and power flow

Integrating multi-sub-microgrid in DC microgrids is able to effectively extend the system's operational life. However, at the same time, it also poses a challenge for system control. Achieving accurate load current sharing between sub-microgrids and minimizing the average voltage fluctuations in the DC microgrids remain the challenging task. In general, DC microgrids typically address the task by using droop control at the primary control and generating the corresponding voltage and current correction terms at the secondary control through the interaction of information within the communication network.

Information flow plays a very significant role in the distributed control mode of DC microgrids. In this paper, the communication network topology is ring-structured. An undirected connected graph $\mathcal{G} = (\Lambda, \mathcal{E})$ can be used to simulate the communication network for a system with n agents, where $\Lambda = \{1, 2, \ldots, n\}$ is the set of agent nodes and $\mathcal{E} \subseteq \Lambda \times \Lambda$ is the edge set. If agent node j can send data to agent node i, it means edge $(i, j) \in \mathcal{E}$ and node j is referred to as node i's neighbor. Given that the graph $\mathcal{G}$ is an undirected connected graph, then edge $(j, i) \in \mathcal{E}$ and node i is also a neighboring node of node j. The adjacency matrix of the graph $\mathcal{G}$ is represented by an $n \times n$ matrix $A = [a_{ij}]$, where $a_{ij} > 0$ only when $(i, j) \in \mathcal{E}$, and $a_{ij} = 0$ otherwise. The weighted degree matrix $D = [d_{ii}]$, where the weighted degree d_{ii} of node N_i is defined as $d_{ii} = \sum_{j=1}^{n} a_{ij}$. With the weighted degree matrix and the adjacency matrix, the Laplacian matrix $L = [l_{ij}] \in \mathbb{R}^{n\times n}$ can be obtained as follows:

$$L = D - A \tag{3}$$

It is well known that in the traditional distributed consensus secondary control strategy, droop control in the primary control plays an important role. The Droop control is based on the predefined droop coefficient R_{vi} :

$$U_i = U_{ref} - R_{vi} I_{bat,i} \tag{4}$$

where U_i, $I_{bat,i}$, U_{ref} are the output voltage, the energy storage battery's output current, and the bus reference voltage of sub-microgrid i, respectively.

However, droop control causes a static bias in the DC bus voltage and current sharing is not achieved with high accuracy due to line impedance mismatch, sensor errors, etc. Thus, the secondary controllers are usually introduced to generate the voltage and the current regulation terms for adjusting the voltage settings of the primary controller by processing information from local and neighboring interactions. The voltage regulation term ΔU_i contains an average voltage observer and a PI controller P_{U_i}, as follows:

$$\begin{cases} \bar{U}_i^t = U_i^t + \int_0^t \sum_{j\in N_i} a_{ij} \left(\bar{U}_j^t - \bar{U}_i^t \right) dt \\ \Delta U_i = P_{U_i} \left(U_{avg}^{ref} - \bar{U}_i^t \right) \end{cases} \tag{5}$$

where $\bar{U}_i^t$ denotes the average observed voltage of sub-microgrid, N_i is node i's neighboring set, $\bar{U}_j^t$ is the average observed voltage of neighbor agent j and U_{avg}^{ref} is the average reference voltage.

The current regulation term ΔI_i used the consensus algorithm with the PI controller P_{I_i} to achieve the aim of current sharing between multiple sub-microgrid.

$$\Delta I_i = P_{I_i} (\sum_{j\in N_i} a_{ij} \left(I_{bat,j}^t - I_{bat,i}^t \right)) \tag{6}$$

In summary, the final voltage reference value for the primary control is given as Eq. (7).

$$u_i^t = U_{avg}^{ref} - R_{vi} I_{bat,i}^t + \Delta U_i + \Delta I_i \tag{7}$$

This typical control strategy divides the controller into the primary control and secondary control, which to a certain extent achieves the voltage regulation and current sharing. According to Eq. (7), it can be seen that this control strategy essentially generates a voltage reference for the primary control of each sub-microgrid. However, this distributed consensus control strategy based on PI controller is more sensitive to the design of model parameters, such as the droop coefficient and the PI controller contained in the two correction terms. In addition, due to the presence of an integral term in the average voltage observation, the integral term causes delay of the microgrids input signal, which is not conducive to the dynamic response of the system, and can lead to large voltage fluctuations under load or PV power fluctuations. This case is not conducive to the long-term operation of the grid.

Inspired by the aforementioned work, this paper leveraged the adaptive capability of reinforcement learning and proposed a MARL algorithm as the secondary controllers for generating the voltage reference values. It eliminated the requirement for model parameter adjustment and reduced the DC microgrids voltage fluctuations.

2.3. Secondary control problem formulation

In this section, a n-player cooperative game problem is formulated. The objective for n players is to pursue the highest global reward through cooperative working. Based on Eqs. (4)–(7), the gap between the training reward and the desired target is mainly determined by two real-time errors, the voltage error and the current error. Voltage error

e_U^t is mainly used to assess the error between the average voltage of the overall microgrid and the reference. Meanwhile, the current error e_I^t is used to assess the error of current sharing, which requires counting the error between the battery current and the average current in each sub-microgrid.

$$e_U^t = \left| \frac{\sum_{i=1}^{n} U_i^t - nU_{avg}^{ref}}{U_{avg}^{ref}} \right| \tag{8}$$

$$e_I^t = \sum_{i=1}^{n} \left| \frac{I_{bat,i}^t - I_{avg}^t}{I_{avg}^t} \right| \tag{9}$$

where $I_{avg}^t = \sum_{i=1}^{n} I_{bat,i}^t / n$ represents the average current at this sampled time ste, which is the average value of the output current of the energy storage battery in each sub-microgrid.

It is worth noting that the two dynamic errors are related to each control agent. Considering that the cooperative game framework is for pursuing the highest global reward, the performance index is as follows:

$$J = \sum_{t=0}^{+\infty} \gamma^t (\alpha e_U^t + \beta e_I^t + \mathcal{H}) \tag{10}$$

where $\gamma \in (0, 1]$ is the discount coefficient and α, β are variable positive weights for different stages depending on the corresponding error, and $\mathcal{H}$ is the excitation when the two dynamic errors step into the expected stage, the rest of the stages are 0.

3. The distributed secondary control strategy

Inspired by the aforementioned work, in this section, we proposed a safe Multi-Agent Proximal Policy Optimization (SMAPPO) reinforcement learning control strategy. For achieving current sharing and voltage regulation, a safety mechanism for the output of the reference voltage is designed in order to prevent dramatic voltage fluctuations. At first, a multi-agent secondary control strategy based on collaborative game theory is established according to the proposed performance index. Then, the implementation of the proposed algorithm is discussed in detail.

3.1. DC microgrids RL control strategy

In our paper, the agent of each sub-microgrid outputs the reference control voltage $\bar{a}_i^t$ to the primary controller. The proposed strategy can keep the system operating steadily and respond quickly to environmental changes through distributed decision-making. According to the collaborative game theory, the control strategy proposed in this paper $(\mathcal{S}, \mathcal{A}, \pi, R, \gamma)$ can be defined as follows: • **Action space** $\mathcal{A}$: In this paper each agent outputs the corresponding action a_i^t, and then according to the safety mechanism used in this paper, the output action a_i^t transformed to the output secondary control signal $bara_i^t$ which is the reference voltage value of the primary control at sampling time t. The standard steady-state average bus voltage in our DC microgrids is denoted by 1 pu and the value is designed as 50V. In this paper, we use continuous actions between 0.8 pu and 1.2 pu. Therefore, the overall action of the whole system is the joint action of each agent, i.e., $\mathcal{A}^t = \bar{a}_1^t \times \bar{a}_2^t \times \cdots \times \bar{a}_n^t$.

• **State space** $\mathcal{S}$: Assuming each agent i can only perceive its own and neighbors' state information. Choose its storage battery's output current $I_{bat,i}^t$, and the output voltage U_i^t as its own state information. According to the general rule in multi-agent collaborative games, the state information of neighboring agents can also be received. Therefore, at each sample time t, the observed state set O_i^t of agent i can be expressed as:

$$O_i^t = [I_{bat,i}^t,\ \{I_{bat,j_1}^t, \ldots, I_{bat,j_N}^t\},\ U_i^t,\ \{U_{j_1}^t, \ldots, U_{j_N}^t\}] \qquad (i, j) \in \mathcal{E} \tag{11}$$

where $\{i, j\} \in \Lambda$ represents the agent i and its neighboring agents j. Our proposed algorithm is centralized training. Thus, the global states information $\mathcal{S}^t$ is used to train the critic network. It can be expressed as Eq. (12) which is the union of each agent's observed state set:

$$\mathcal{S}^t = O_1^t \cup O_2^t \cup \cdots \cup O_n^t \tag{12}$$

• **Transition probability** π: At sample time t, under the joint action A^t of multiple agents, DC microgrids will transition from state $\mathcal{S}^t$ to $\mathcal{S}^{t+1}$ with a transition probability $P_S(\mathcal{S}^{t+1} | \mathcal{S}^t, A^t)$. For each sub-microgrid, its observation space will transition from state O_i^t to O_i^{t+1} with a transition probability function $\pi = P_O(O_i^{t+1} | O_i^t, a_i^t)$. The proposed method does not need any prior knowledge of models or transition probabilities at the secondary control layer of each sub-microgrid. In addition, in the primary layer the commonly used method of double PI is adopted, similar to [31].

• **Reward function** $\mathcal{R}$: It is well known that the two main objectives of DC microgrids secondary control are voltage regulation and current sharing. Based on the analysis in Section 2, it can be seen that both objectives are ultimately achieved by giving a reference voltage for primary control. Thus, it has a strong coupling relationship. In the training process, agents are used to explore good control strategies through continuous trial, but the convergence speed of the training is greatly affected by the coupling effect of the above two main objectives in this process.

Algorithm 1 reward function

Initial the time threshold τ, the average reference voltage U_{avg}^{ref} and the coefficient $\alpha_1, \alpha_2, \beta_1, \beta_2$.
REPEAT
For the Step starts from $t = 1$ to MaxStep execute
Obtain the global state S
Calculate the errors of voltage e_u^t and current e_I^t at the sample time t.

```
If e_U^t <= 0.05
    If count <= τ
        count = count + 1
        R = −α1 ∗ e_U^t − β1 ∗ e_I^t
    else
        #into second stage
        count = count
        If e_I^t <= 0.05
            R = α2 ∗ (0.05 − e_U^t) + β2 ∗ (0.5 − e_I^t) + 0.5
        elseif e_I^t <= 0.5
            R = α2 ∗ (0.05 − e_U^t) + β2 ∗ (0.5 − e_I^t)
        else
            R = α2 ∗ (0.05 − e_U^t) − β1 ∗ e_I^t
elseif e_U^t <= 0.2
    R = −α1 ∗ e_U^t − β1 ∗ e_I^t
    count = count − 2
else
    R = −50
    count = 0
return R
```

Therefore, according to the performance index (10), this paper transforms it into a two-stage reward function to reduce the coupling effect of the two main objectives to a certain extent and improve the convergence performance of the proposed method by adjusting the emphasis on different control objectives under each stage. In the first stage, the reward function mainly focused on quickly converging the DC bus voltage to the safe average bus voltage with the weighting coefficient α_1 is set to 8, and the current sharing is as a secondary objective with the weighting coefficient β_1 is set to 0.5. Similar to [32], the reward conditions are divided into three categories based on the difference between the output bus voltage and the reference voltage: normal zone ([0.95 pu, 1.05 pu]), violation zone ([0.8 pu, 0.95 pu] ∪

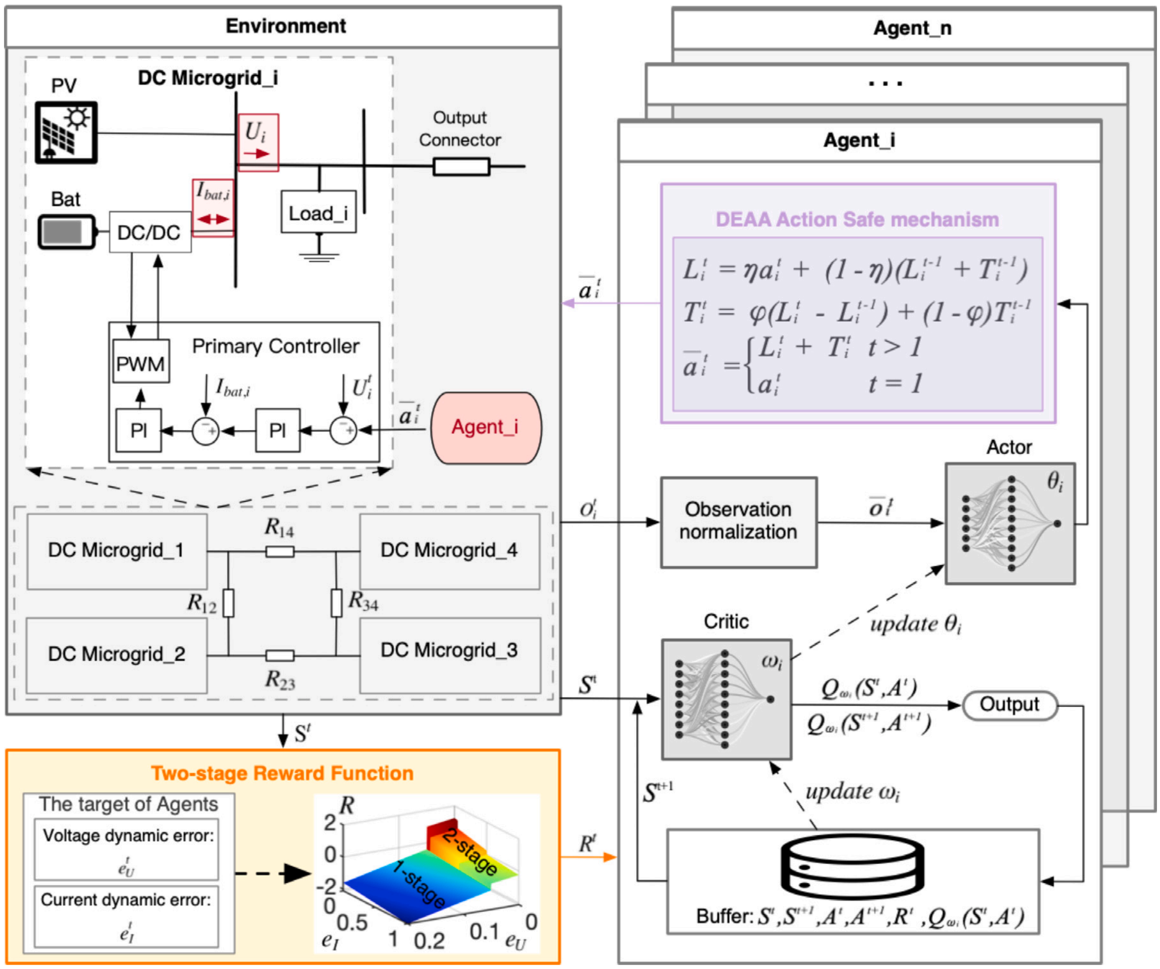


Fig. 2. The secondary control framework for DC microgrids based on the proposed method, which incorporates the interaction between the environment and the agents. Each agent outputs the reference voltage at the sample time by the local observed states of voltage and storage battery current. Then the voltage error and the current error are calculated based on the global state to obtain the global reward, which used to evaluate the whole control strategy.

[1.05 pu, 1.2 pu]), and diverging zone in all other cases. At the same time, in order to ensure that the agents have been able to maintain a high probability of searching within the safe voltage error range of 5% rather than accident, we set a time threshold τ. Only if the agents maintain explore within a safe voltage range beyond the time threshold τ, the reward function enters the second stage of the reward function.

In the second stage, the emphasis of the two main objectives is adjusted by changing their respective weights in the exploration process of the agents, which are modified to $\alpha_2 = 5$ and $\beta_2 = 1$, respectively. At the same time, the overall reward will be shifted from negative to positive, encouraging the agents to step directly into the safe operating range during future exploration and avoid stepping out of it. In addition, to enable the agent to learn our desired control strategy, when the agent outputs our desired control strategy, an incentive mechanism is introduced to give the agent a constant positive reward 0.5. This value means that when the current error is less than 5% and the average voltage is also within the safe operation range.

The reward function in detail is as Algorithm 1 shows. Afterward, the agents update their policy based on the global reward R, establishing the mapping from each observed state O_i^t to the action and guiding them to a better policy.

• **Discount factor** γ: the discount factor γ establishes the relative importance of immediate reward and future rewards. In this paper, each agent is trained to output a voltage reference value to the primary control, the expected control performance is mainly determined by the short-term dynamic changes of the voltage and the magnitude of the error between currents. In this paper, the γ is set as 0.8.

3.2. Training and execution of the proposed strategy

As shown in Fig. 2, each agent has adopted offline centralized training before online distributed execution. During the training phase, each agent continuously improves its collaborative abilities to explore the higher global rewards R. And during the execution phase, each agent only needs to take action at the control step with its own observation state information without global information. The algorithm is implemented through the following two processes: (1) state normalization and the safe mechanism design; (2) network parameters updating. The specific implements' details are as follows:

3.2.1. State normalization and the safe mechanism design

In this paper, the proposed Safe Multi-Agent Proximal Policy Optimization (SMAPPO) algorithm is an extension of the PPO algorithm. Besides, the input observation states normalization and the safety mechanism for the output actions are designed to reduce the possibility of the system breaking down due to dramatic fluctuations in the output actions.

At first, for secondary control of the multi-agent DC microgrids, each agent needs to make decisions based on its own and neighbors' information. Therefore, each agent needs to establish communication channels with its neighbor agents and exchange necessary information that is advantageous for the system control. In this paper, the output voltage U_i^t and the output storage battery current $I_{bat,i}^t$ are chosen as the local information of agent i. According to the graph theory described in

Section 2, the information of agent j with edge$(i,j) \in \mathcal{E}$ is considered as the neighbor information of i. Since the voltage observation values and current observation values have different characteristic ranges, which may cause bias in high-value features and make the network weight parameters explode, feature normalization is performed according to Eq. (13) to ensure the influence of each feature.

$$\bar{o}_i^t = \frac{O_i^t - mean(O_i^t)}{std(O_i^t)} \tag{13}$$

In addition, it is noted that the proposed algorithm in this paper is based on PPO, an on-policy MARL algorithm that may choose a random action according to a normal distribution probability at each time step. This may result in noise fluctuations being sampled into the action sequence even after the algorithm has converged and cause unnecessary fluctuations for the whole system. Therefore, As a result, processing the output actions is necessary. The most common method is to limit them, however, this may result in missing some system dynamic performance when the system fluctuates greatly. In this paper, to solve the problem, the action sampled at each time step is secured by the Double Exponential Averaging Algorithm (DEAA). According to the forward-done actions' horizontal component L^{t-1} and trend component T^{t-1}, set two safe factors η and φ to get the safe action $\bar{a}_i^t$ at this step, as follows:

$$\begin{aligned} L^t &= \eta a_i^t + (1-\eta)\left[L^{t-1} + T^{t-1}\right] \\ T^t &= \varphi\left[L^t - L^{t-1}\right] + (1-\varphi)T^{t-1} \\ \bar{a}_i^t &= \begin{cases} L^t + T^t & t > 1 \\ a_i^t & t = 1 \end{cases} \end{aligned} \tag{14}$$

where $\eta \in [0,1]$ is the horizontal hyperparameter, $\varphi \in [0,1]$ is the trend hyperparameter and a_i^t is the action taken by the agent i at time t. When $\eta = 1$ represents the L^t is the original action sampled by the agent i and without safety guarantees. The more η trend to 0, the smaller the L^t changed, and if $\eta = 0$ means the chosen action is unchanged which keeps the initial value all time. And for φ, it influences the action's rate of change. The closer the φ is to 1, the better the trend following between $\bar{a}_i^t$ and the agent's sample outputs, which could adjust the output dynamic characteristic of $\bar{a}_i^t$. At the initial time step $t = 1$, set the $L^0 = U_{avg}^{ref}$ and $T^0 = 0$. In this paper, $\eta = 0.5$ and $\varphi = 0.2$ are adjusted by cross-validations and set a constant value for each one before training.

3.2.2. Networks parameters updating

The proposed algorithm is used centralized training to update the network parameters to obtain the desired control strategy and then used the distributed execution that can successfully prevent system instability brought on by a SPoF and fully utilize the benefits of distributed control. We use the agent i as an example to discuss how to centralize training and execute distributedly of the proposed method.

In the centralized training phase, the critic network serves as the central evaluator by collecting global information while the actor network only uses local and neighbor information to sample the action. Then, obtain the global reward R^t from the DC microgrids system. For the secondary control goals of DC microgrids, our proposed algorithm is aimed to achieve the maximum global reward R. Therefore, the critic network needs to generate a centralized state function $V_{\omega_i}\left(S_t\right)$, to evaluate the actions from the global perspective and prompt the agents to choose better actions. Then the critic network optimizes its parameters ω_i by minimizing the following loss function:

$$\begin{aligned} L_{\omega_i} = E_D \Big\{ \max \Big[& \left(V_{\omega_i}\left(S^t\right) - R_t - \gamma V_{\omega_i}\left(S^{t+1}\right)\right)^2, \\ & \Big(\text{clip}\Big(V_{\omega_i}\left(S^t\right), V_{\omega_i}\left(S^t\right) - \epsilon \\ & V_{\omega_i}\left(S^t\right) + \epsilon\Big) - R_t - \gamma V_{\omega_i}\left(S^{t+1}\right)\Big)^2 \Big] \Big\} \end{aligned} \tag{15}$$

where $\mathcal{D}$ represents the minibatch, S^{t+1} is the next sample step global states and ϵ is a clip fraction, which is predetermined as 0.2. The clip fraction is able to avoid the excessive modification of the loss function L_{ω_i}.

On the other hand, for the actor network, the parameters θ_i and the output of the sampled action are both related to the current observation state $\bar{o}_i{}^t$ and the state function $V_{\omega_i}\left(S^t\right)$ output by the critic network. The actor network's optimization first-order objective function is to maximize the loss function L_{θ_i}, as illustrated below:

$$\begin{aligned} L_{\theta_i} = E_D \Big\{ & \min\left(\kappa_t\left(\theta_i\right), \text{clip}\left(\kappa_t\left(\theta_i\right), 1-\epsilon, 1+\epsilon\right)\right) \hat{A}_t^{GAE(\gamma,\zeta)} \\ & + \phi\mathcal{E}\left[\pi_{\theta_i}\right]\left(\bar{o}_i\right) \Big\} \end{aligned} \tag{16}$$

where $\kappa_t(\theta) = \pi_{\theta_i}(\bar{a}_i^t|\bar{o}_i^t) \,/\, \pi_{\theta_i}^{old}(\bar{a}_i^t|\bar{o}_i^t)$ is the difference ratio between the new policy and old policy, ϕ is the entropy coefficient hyperparameter and $\mathcal{E}$ is the policy entropy. The entropy bonus is the output of a neural network combined the policy π_{θ_i} with the state function $V_{\omega_i}\left(\bar{o}_i\right)$ and can enhance the agent i's exploration ability. $\hat{A}_t{}^{GAE(\gamma,\zeta)}$ is the generalized advantage estimator (GAE), as follows:

$$\hat{A}_t{}^{GAE(\gamma,\zeta)} = \sum_{l=t}^{T-1} (\gamma\zeta)\delta_l \tag{17}$$

where the ζ is the smoothing factor for GAE and $\delta_t = r_i^t + \gamma V_{\omega_i'}\left(\bar{o}_i{}^{t+1}\right) - V_{\omega_i}\left(\bar{o}_i{}^t\right)$.

In the distributed execution phase, the critic network is dropped and only the actor network takes part in online execution. This means that the well-trained agent no longer needs to collect global state information but merely needs the observation states to achieve secondary control objectives of the DC microgrids. This not only eases communication across subnets but also enables real-time mapping of control variables through the network structure, saving time and resources while greatly enhancing the system's dynamic performance.

Table 2
System parameters.

Parameter	Value
Reference average voltage (U_{avg}^{ref}/V)	50
Line resistance ($R_{12}, R_{23}, R_{43}, R_{14}/\Omega$)	0.3, 0.5, 0.7, 0.5
Battery voltage (U_{bat}/V)	24
Battery rated capacity (C_{bat}/Ah)	50
Primary control voltage loop	$k_p^U = 16, k_i^U = 1800$
Primary control current loop	$k_p^I = 0.043, k_i^I = 0.65$

4. Experimental results

In this section, we will introduce the experiment setup of the hardware-in-the-loop (HIL) platform, and then conduct experiment results. The superiority of the proposed method will be verified by comparing with existing methods.

4.1. Experiment setup

The studied DC microgrids topology consists of four sub-microgrid connections, taking into consideration the various line impedances existent. The control topology of a sub-microgrid consists of a primary controller and the proposed secondary reinforcement learning agent.

4.1.1. Parameter setting

The system parameters are listed in Table 2, which are obtained by adjusting the control effect during the experiment. To validate the proposed SMAPPO, four agents are employed in the experiment, and their parameters are specified in Table 3 (the parameters of each agent are set to be identical). These hyperparameter settings can be obtained from the original MAPPO baselines implementation [29]. The DC microgrid experiment platform is created based on MATLAB 2022b and the training process is conducted on CPU T7700 2.4 GHz.

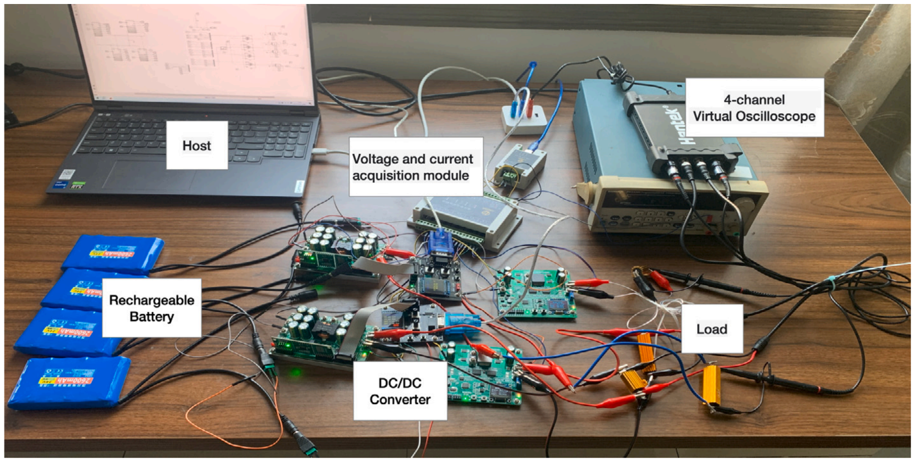


Fig. 3. Hardware-in-loop experiment platform.

Table 3
Parameters setting of the proposed SMAPPO.

Parameter	Value
Clip fraction (ϵ)	0.2
Entropy coefficient (ϕ)	0.01
Smooth factor of GAE (ζ)	0.95
Sample step time (T_s)	0.001
Mini batch size	168
Max step per episode	400
Learning rate for critic network	$1e-4$
Learning rate for actor network	1.5e−4

4.1.2. Hardware setup

In order to evaluate the effectiveness of the proposed SMAPPO control strategy, this paper carries out the experimental verification in a hardware-in-loop (HIL) experimental platform. The experimental platform is shown in Fig. 3, which includes the host computer, the 4-channel virtual oscilloscope, four DC/DC converters along with the current and voltage acquisition module. In this HIL platform, the host computer is used to configure multiple agent controllers and simulate PV generation equipments using MATLAB. The current and voltage acquisition module ZS-9VCI-485-CAN is used to collect the agents' observation status information, while the 4-channel virtual oscilloscope is used to monitor the system status and display the experimental results.

The communication topology of the HIL experimental platform uses an undirected connection graph same as the physical connection graph, which is a ring structure with a specific adjacency matrix $A = [0,1,0,1;1,0,1,0;0,1,0,1;1,0,1,0]$. Real-time information interaction between the simulated PV, the controller and the actual battery control object was achieved by using the serial port with a baud rate of 115 200. Adopts 10 ms as the communication rate for HIL experiments, which is in accordance with the IEEE standard.

4.2. Algorithm training performance

In this subsection, the training conditions of the agents in this paper are described firstly, and then the training performance is compared to the proposed SMAPPO without state normalization, without action safety mechanism, and a one-stage reward function, and shows the improvement of the proposed algorithm in training efficiency and control performance. Compared to the proposed method with Multi-agent Deep Deterministic Policy Gradient (MADDPG), the trained reward shows that the proposed method is better able to achieve the two main objectives of secondary control of DC microgrids.

4.2.1. Training conditions

The SMAPPO training process is the process of maximizing the global accumulated rewards by constantly interacting with the environment and taking optimal secondary control decisions in different environmental states. In this process, the training conditions play a significant role. The quality and diversity of training conditions are important guarantees for its generalization and superior performance.

In DC microgrid secondary control, it is mainly affected by the renewable energy generation power and the load demand power. Therefore, in order to ensure the diversity of the training conditions, the source-load power changes are constructed as shown in the Table 4, where Γ represents a random number between 0 and 0.2. In addition, considering the real-time random fluctuation of the source-load power, Gaussian noise with a mean value of 0 and a standard deviation of 100 W is introduced to further improve the quality of the training conditions and to guarantee the generalization ability of the multi-agent.

4.2.2. Algorithm convergence analysis

In order to improve the training efficiency of the algorithm, our proposed algorithm integrates three modules, the input observation state normalization, the output action safe mechanism and a two-stage reward function. In particular, the input observation state normalization is primarily used to avoid the influence of the difference in the change dimension. The output action safe mechanism is used to optimize the action output and filter out inappropriate actions obtained from low probability sampling. And the two-stage reward function is used to reduce the coupling effect of the two secondary control objectives on the training of the algorithm. In Fig. 4, it can be seen that how the action safe mechanism and the input observation state normalization impacts the algorithm convergence performance. Fig. 5 demonstrates the two-stage reward function's effectiveness. In general, normalizing the input observation states, using the output action safe mechanism and the designed two-stage reward function are all beneficial to the proposed algorithm performance.

Additionally, the proposed SMAPPO algorithm is compared with the widely used MADDPG algorithm. MADDPG also adopts a centralized training and distributed execution framework. As depicted in Fig. 6, the experimental result reveals that the SMAPPO algorithm is superior to MADDPG. This is due to the fact that the proposed SMAPPO algorithm is derived from the PPO framework and includes action entropy incentives to improve agents' exploration capabilities and help them search for better control strategies. As a result, SMAPPO can produce a better control strategy for the DC microgrids.

Fig. 7 shows the control performance of SMAPPO and MADDPG during the online execution stage. We executed the same test condition

Table 4
Training conditons.

First phase	Second phase	Third phase
$0 \sim 0.2 + \Gamma(s)$ 700 W; 700 W; 550 W; 550 W	$0.2 + \Gamma \sim 0.6 + \Gamma(s)$ 200 W; −300 W; −400 W; −500 W	$0.6 + \Gamma \sim 1(s)$ 300 W; 500 W; 500 W; 200 W

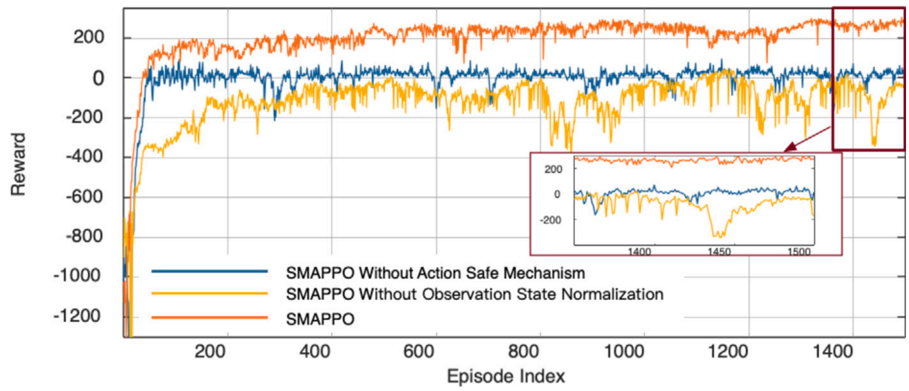


Fig. 4. The training reward comparison of SMAPPO without action safety, SMAPPO without state normalization, and the proposed SMAPPO.

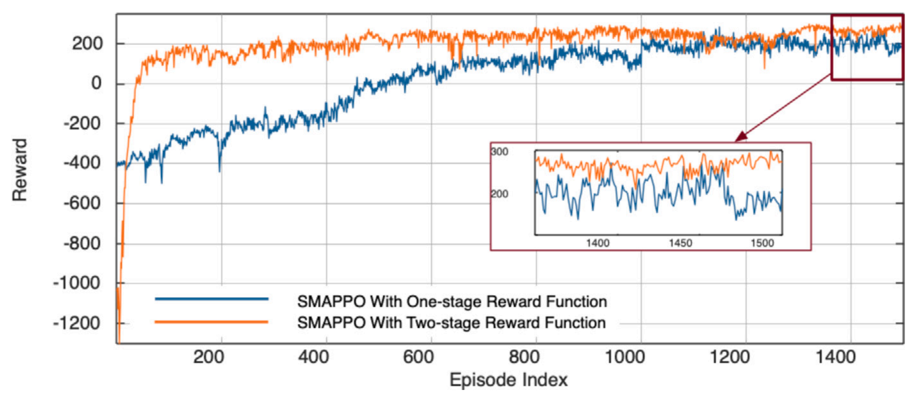


Fig. 5. The training reward comparison of one-stage reward function and two-stage reward function.

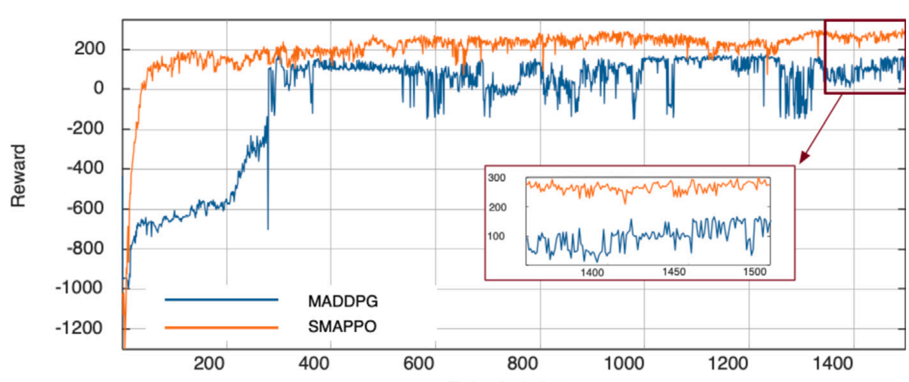
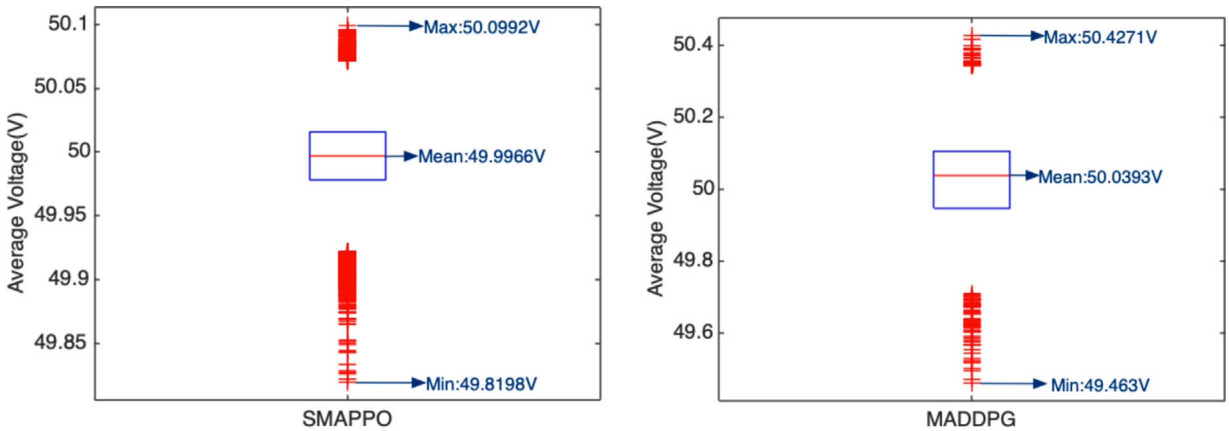
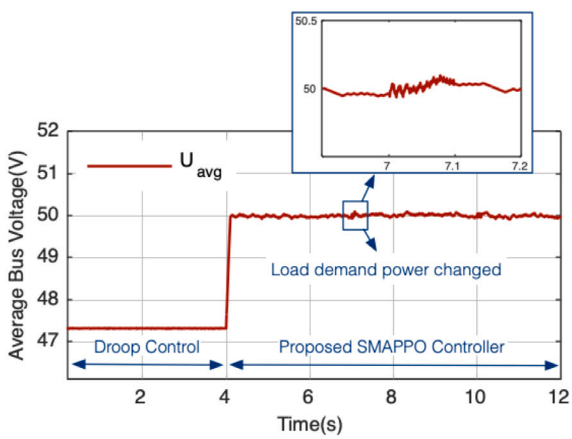


Fig. 6. The training reward comparison of MADDPG and the proposed SMAPPO.

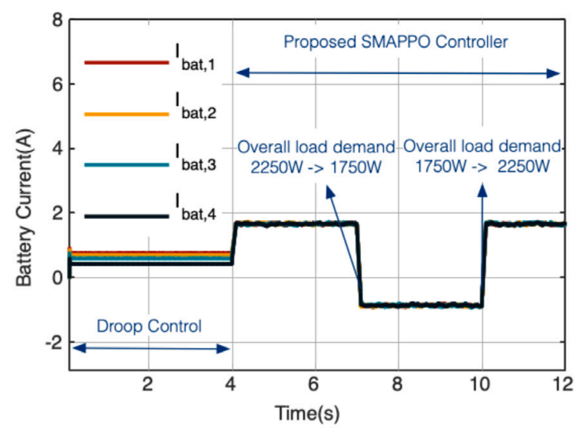


(a) Control performance of SMAPPO's online execution (b) Control performance of MAPPDG's online execution

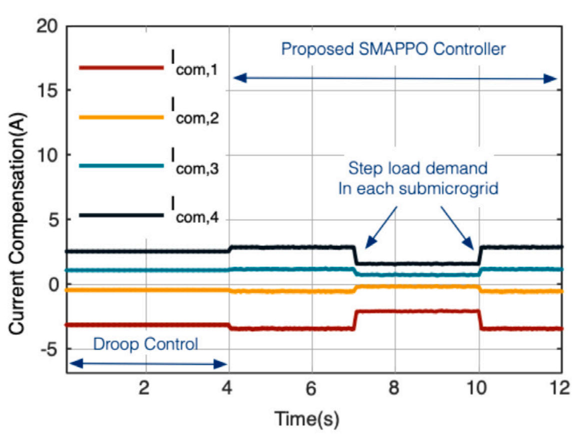
Fig. 7. Algorithm stability analysis.



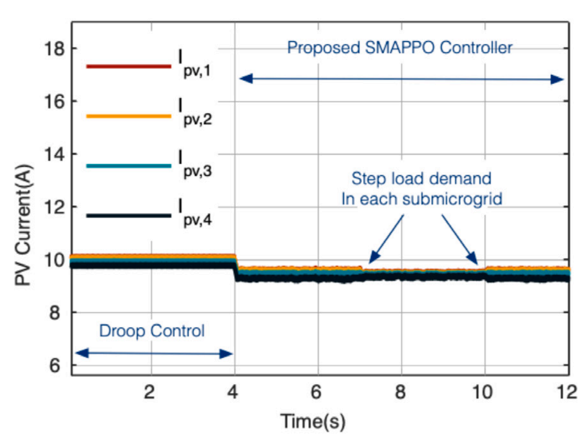
(a) Average bus voltage



(b) Storage batteries output current



(c) Current compensation of each sub-microgrid



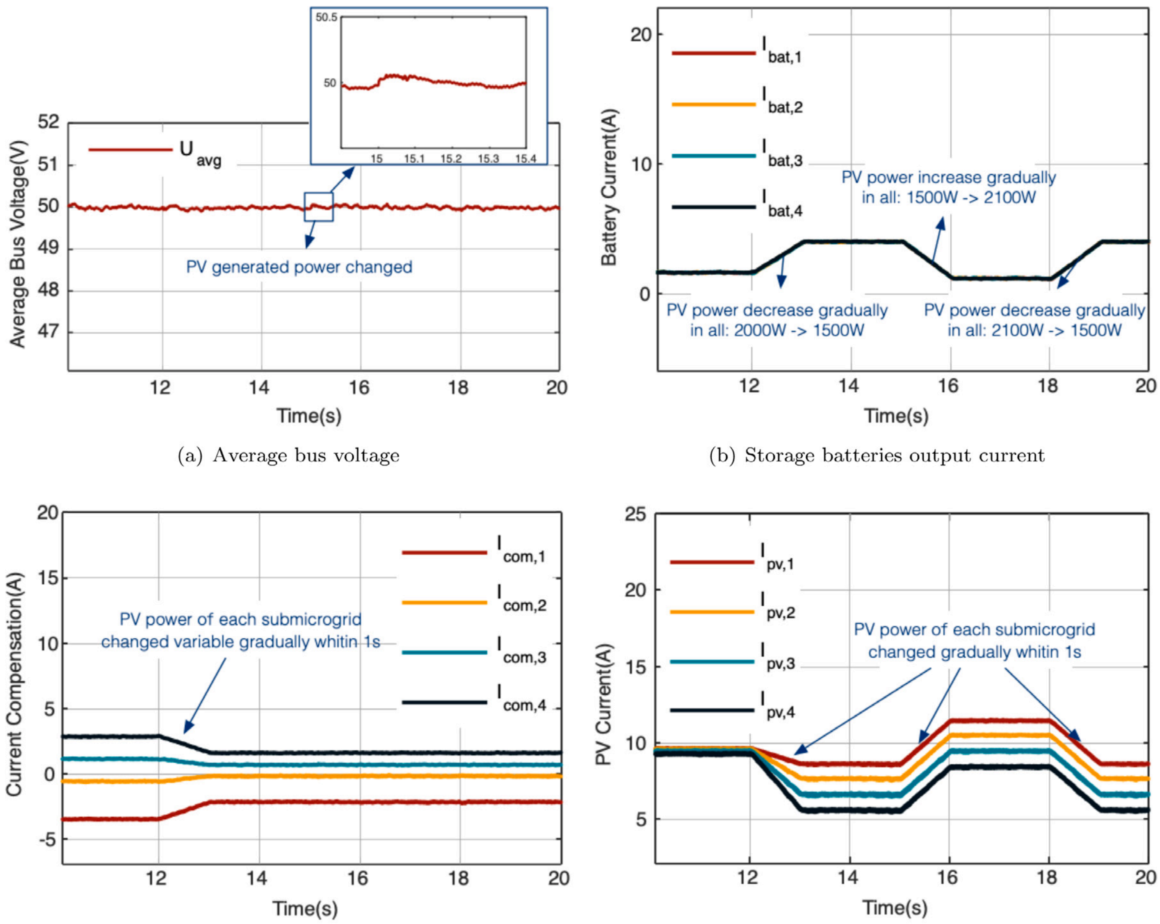
(d) PVs output current

Fig. 8. Experimental results of Case I: the load demand power change.

online for 500 rounds respectively and recorded the control of the average bus voltage in each round during the execution stage.SMAPPO is significantly better than MADDPG. In these rounds, the average bus voltage under the control strategy of SMAPPO is 50.0992 V at maximum and 49.8198 V at minimum, which has a good stability.

4.3. Control performance of the proposed SMAPPO

In this section, two case experiment experiments are conducted to verify the effectiveness of the proposed SMAPPO algorithm, involving sudden load changes and PV generation variations, respectively. And



(a) Average bus voltage

(b) Storage batteries output current

(c) Current compensation of each sub-microgrid

(d) PVs output current

Fig. 9. Experimental results of Case II: PV generated power change.

then compared the control performance of the distributed consensus method, MADDPG and the proposed SMAPPO.

4.3.1. Case I – Load power change

In this case, the main focus of the study is on the control performance of the proposed SMAPPO during sudden load changes. The generated power from PVs in every subsystem stayed constant throughout the case experiment. Each PVs generated power worked at the MPPT mode. As shown in Fig. 8, the case I of the experiment can be divided into four stages as follows:

• **Stage** 1 (0∼4 s): the proposed SMAPPO secondary control agents are not activated in the control and the primary controller of each subsystem is controlled by the traditional droop control. The load demand power of each sub-microgrid are 750 W, 600 W, 500 W, and 400 W.

• **Stage** 2 (4∼7 s): at t = 4 s the proposed SMAPPO secondary control agents are activated and the droop control is discarded. The agents give the reference voltage signal to each own primary controller.

• **Stage** 3 (7∼10 s): the load demand power of each sub-microgrid decreased by 200 W, 150 W, 100 W, 50 W respectively. That means the overall load demand power decreased by 500 W.

• **Stage** 4 (10∼12 s): at t = 10 s, the load demand power of each sub-microgrid increased by 200 W, 150 W, 100 W, 50 W, respectively. That means the overall load demand power increased by 500 W.

As can be seen in Fig. 8, within stage 1, utilizing the droop control cannot achieve the targets of voltage regulation and current sharing well. In particular, the average bus voltage has a large deviation from the reference value. Upon entering stage 2, the droop control strategy is removed and activates the proposed SMAPPO secondary control agents. Through the interaction of its own information with its neighbors, the effects of the line impedance are eliminated, realizing the current sharing and voltage regulation. In addition, in stages 3 and 4, scenarios involving sudden load demand power increased and reductions were respectively simulated, and the average bus voltage fluctuations were slight and remained at the reference value. Since in stage 3, the overall load power required is less than the PV generated power, the energy storage batteries in each sub-microgrid are charged with the same current. And in stage 4, as the load demand power rises, the energy storage batteries are again discharged with the same current.

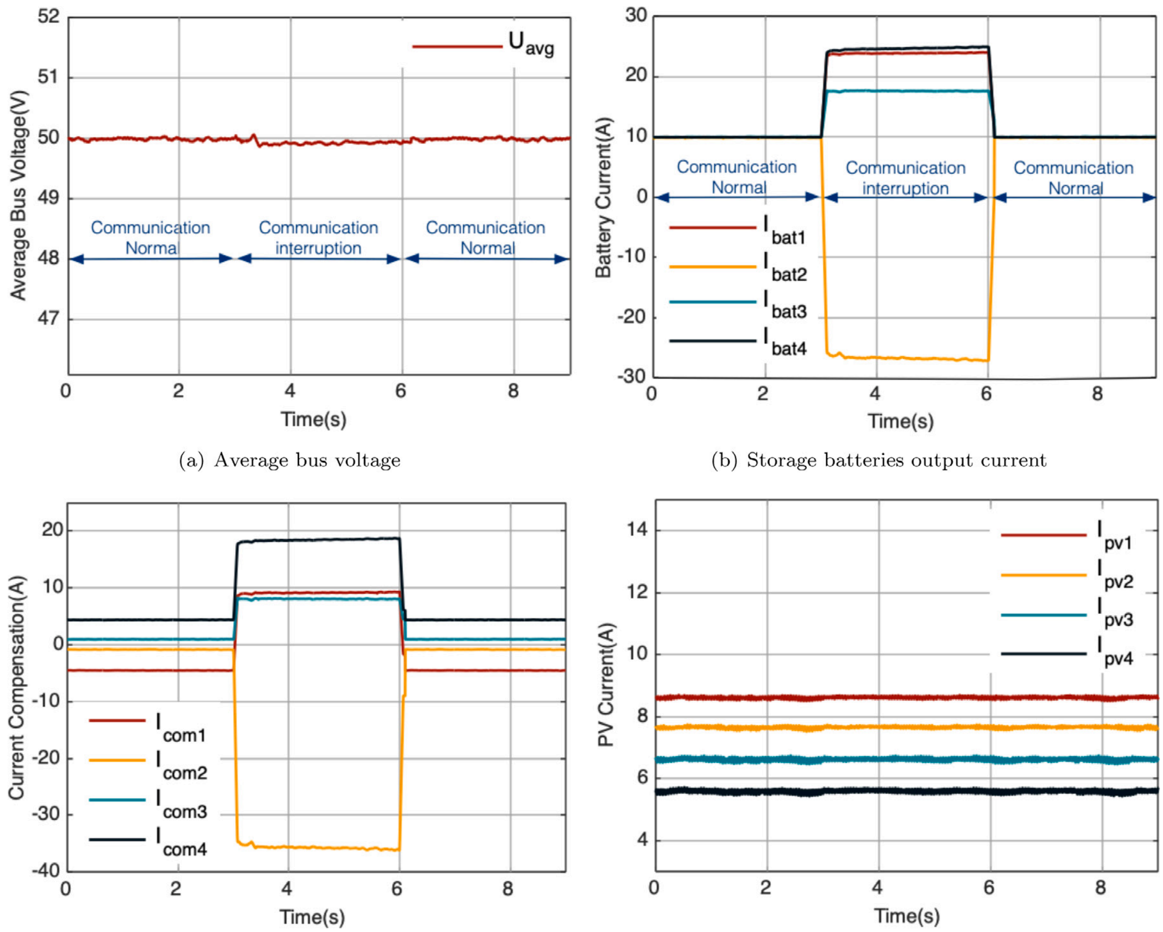
From the above experimental results, it can be seen that the proposed SMAPPO method can eliminate the impact of line impedance inconsistencies on the current sharing and quickly maintain the stability of the average bus voltage during sudden load demand power changes, ensuring that the system operates safely.

4.3.2. Case II – PV generation power change

In this case, the main focus of the study is on the control performance of the proposed SMAPPO during PV generation power changes. The overall load power kept 2250 W and the load power of each sub-microgrid stayed constant at 750 W, 600 W, 500 W, 400 W, respectively. As shown in Fig. 9, the case experiment can be divided into four stages as follows:

• **Stage** 1 (10∼12 s): during this time, each PV generation power of the sub-microgrid worked at the MPPT model and kept the same at 500 W.

• **Stage** 2 (12∼15 s): at t = 12 s, the PV generated power by each sub-microgrid started to vary. And between 12 to 13 s, the respective



(a) Average bus voltage

(b) Storage batteries output current

(c) Current compensation of each sub-microgrid

(d) PVs output current

Fig. 10. Experimental results of Case III: PV generated power change.

power generated gradually decreased to 450 W, 400 W, 350 W, 300 W. Then the PV generated power kept constant for two seconds.

• **Stage** 3 (15∼18 s): at $t = 15$ s, the PV generated power by each sub-microgrid started to increase. And between 15 to 16 s, the respective power generated gradually increased to 600 W, 550 W, 500 W, 450 W. Next, the PV generated power by each sub-microgrid kept constant for two seconds.

• **Stage** 4 (18∼20 s): at $t = 18$ s, the PV generated power by each sub-microgrid started to decrease. And between 18 to 19 s, the respective power generated gradually decreased to 450 W, 400 W, 350 W, 300 W. Then the PV generated power kept constant for one second.

The experiment experimental results of this case are shown in Fig. 9. As can be seen from either stage 1 or stage 2, the proposed SMAPPO can also achieve coordinated energy control whether the PVs generation power are the same or different. After that, regardless of whether the PV generation power increased in stage 3 or reduced in stage 4, the average bus voltage fluctuates slightly and can be maintained at the rated value. During the two stages, each sub-microgrid's energy storage battery current keeps sharing while the PV generation power changes. In addition, it can be seen from Fig. 9 that with the linear change of PV generated power, the proposed method also shows good dynamic performance.

From the experiment experimental results of case II, it can be seen that our proposed SMAPPO control strategy is still able to share the load current and maintain the average bus voltage at the rated value. The system can be operated safely and stably with good control performance despite the increase or reduction of the PV generated power.

4.3.3. Case III – Communication interruption and restoration

In this case, the study focuses on whether the proposed SMAPPO is still able to achieve the DC microgrid secondary control objectives when the communication is restored to normal after a communication interruption. Therefore, in this case, the load power of each sub-microgrid is always maintained at 750 W, 600 W, 500 W, 400 W, and the PV generated power is always maintained at 450 W, 400 W, 350 W, and 300 W. As shown in Fig. 10, the experiment can be divided into the following three phases:

• **Stage** 1 (0∼3 s): during this time, each sub-microgrid is in a normal operating state and the communication links for information interaction are working normally.

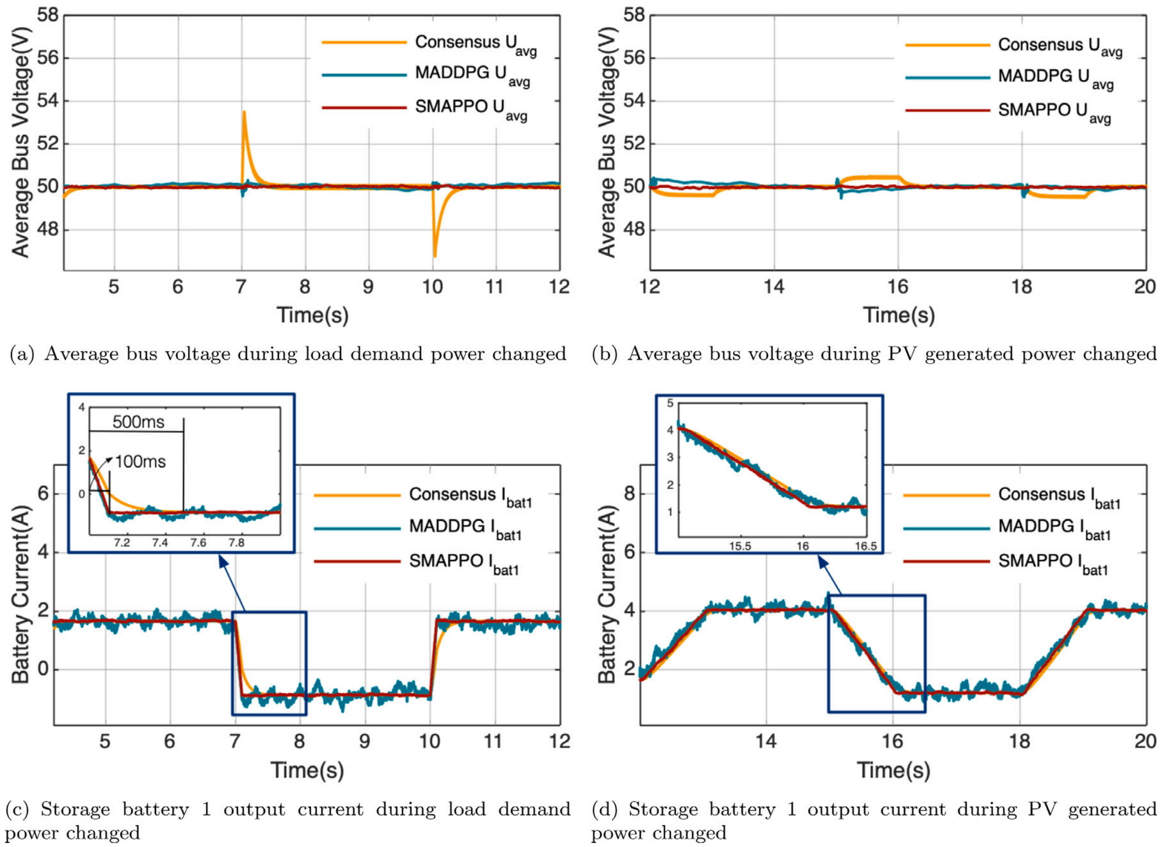
• **Stage** 2 (3∼6 s): at $t = 3$ s, the communication link of sub-microgrid 2 is interrupted.

• **Stage** 3 (6∼9 s): at $t = 6$ s, the communication link of sub-microgrid 2 restored normal operation

The experimental results of this case are shown in Fig. 10. As can be seen from stage 1, the SMAPPO proposed in this paper is able to maintain the average bus voltage at the rated value and keep the energy storage battery current sharing among the sub-microgrids. In addition, from stage 2 to stage 3, it can be seen that the SMAPPO strategy proposed in this paper can still achieve the secondary control objectives among sub-microgrids after the communication interruption is restored, with high stability and reliability.

4.3.4. Comparison with existing methods

In this case, the proposed method is compared with the traditional distributed consensus method and the widely used MADDPG method



(a) Average bus voltage during load demand power changed (b) Average bus voltage during PV generated power changed

(c) Storage battery 1 output current during load demand power changed (d) Storage battery 1 output current during PV generated power changed

Fig. 11. The experimental results comparison of the traditional distributed consensus method, MADDPG and the proposed SMAPPO.

under the same conditions, that is, the same load demand power profile and PV generated power profile. The initial load demand power are 750 W, 600 W, 500 W, 300 W, respectively. The initial PV generated power of all sub-microgrids are both 500 W. The rated average bus voltage is set to 50 V. The experiment process of each method is divided into the following six stages:

- **Stage** 1 (4∼7 s): the system worked in the initial situation.
- **Stage** 2 (7∼10 s): the load demand power of each sub-microgrid decreased by 200 W, 150 W, 100 W, 50 W respectively. And the PV generated power kept constant.
- **Stage** 3 (10∼12 s): in this case, the load demand restored to 750 W, 600 W, 500 W, 300 W, respectively. And each PV generation power of the sub-microgrid also remained unchanged.
- **Stage** 4 (12∼15 s): in the second between 12 to 13 s, the respective power generated gradually decreased to 450 W, 400 W, 350 W, 300 W. Then the PV generated power kept constant for two seconds. And the load demand power kept constant.
- **Stage** 5 (15∼18 s): between 15 to 16 s, the respective power generated gradually increased to 600 W, 550 W, 500 W, 450 W. The PV generated power by each sub-microgrid keeps constant for two seconds. And the load demand power kept constant.
- **Stage** 6 (18∼20 s): between 18 to 19 s, the respective power generated gradually increased to 450 W, 400 W, 350 W, 300 W. Then the PV generated power kept constant for one second. And the load demand power kept constant.

Fig. 11 and Table 5 show the experiment results compared with other advanced methods. From Table 5 and Figs. 11(a) and 11(b), it can be seen that the control strategy based on the traditional distributed consensus method has the highest average voltage fluctuations and longer stabilization time during sudden load demand power changes. Compared to the reference voltage value, it is found that the positive maximum error reaches 6.9974%, and the negative maximum error also has 6.4904%. In addition, the system controlled by the MADDPG-based

Table 5
Positive max error, negative max error, mean error and standard error comparison of proposed SMAPPPO, MADDPG and the traditional distributed consensus method.

Case	Method	Pos. max error (%)	Neg. max error (%)	Mean error (%)	Std error
Load change (4 s∼12 s)	Consensus	6.9974	6.4904	0.0124	0.4276
	MADDPG	0.6094	0.8616	0.0982	0.083
	SMAPPO	**0.1884**	**0.152**	**0.0068**	**0.0302**
PV change (12 s∼20 s)	Consensus	0.9968	1.0234	0.0944	0.2458
	MADDPG	0.8542	1.074	0.0304	0.1293
	SMAPPO	**0.1502**	**0.2062**	**0.0296**	**0.03**

strategy has the positive maximum error of 0.6094% and the negative maximum error of 0.8616% compared to the reference voltage. In contrast, using the SMAPPO control strategy proposed in this paper, the positive maximum error is only 0.1884% and the negative maximum error is 0.152% compared to the reference voltage. During the PV ramp variation, the traditional distributed consensus control strategy cannot even return the average reference voltage, and the MADDPG-based control strategy has the positive maximum error of 0.8542% and the negative maximum error of 1.074%, whereas the proposed SMAPPO control strategy has the positive maximum error of only 0.1502% and the negative maximum error of 0.2062% compared to the reference voltage. All these show that the proposed SMAPPO has better dynamic response performance.

For the battery current sharing, the output current of bat1 is chosen as an example since the currents are generally consistent between each battery. As can be seen in 11(c) and 11(d), the traditional distributed consensus control strategy needs take 550 ms to achieve current sharing at $t = 7$ s during load demand power step change. During the PV ramp change phase, the current sharing rate performs well with a slight lag. For the MADDPG-based control strategy, despite being able to respond dynamically faster to load demand power or PV generated power change, it has large current fluctuations in the steady state due to it falling into a local optimum in the training process. And the method proposed in this paper is able to respond faster to load fluctuations, achieving current sharing at 100 ms at $t = 7$ s, and also has better following performance when the PV ramp changes.

The experiment results of Case IV show that the proposed method has the following advantages: (1) the proposed method has good dynamic performance and is more robust to load demand power and PV generated power fluctuations; (2) it can achieve current sharing at a faster speed.

5. Conclusion

This paper proposed a two-stage multi-agent reinforcement learning secondary control method for DC microgrids with the aim of smoothing the DC bus fluctuations and accelerating the convergence time. The proposed algorithm is trained in the centralized way, and executed in a distributed fashion, which does not require accurate model parameters of the DC microgrids. Compared with the distributed consensus strategy, the strategy proposed in this paper improves the recovery of the current sharing rate of each sub-microgrid by 450 ms. And compared with the MADDPG-based secondary control strategy for DC microgrids, the strategy proposed in this paper has better stability in dealing with system fluctuations, with the maximum error from the rated reference value of the average bus voltage being less than 0.2062%.

In future, it is deserve to further accelerate the response speed of bus voltage adjustment when achieving the distributed energy coordination among DC Microgrids. Meanwhile, it is promising to extend the proposed RL-based secondary control strategy to AC Microgrids by integrating the AC electrical characteristics, such as the phase angle and reactive power.

CRediT authorship contribution statement

Fei Li: Supervision, Project administration. **Weifei Tu:** Writing – original draft, Software, Investigation. **Yun Zhou:** Supervision, Project administration. **Heng Li:** Writing – review & editing, Formal analysis. **Feng Zhou:** Writing – review & editing, Data curation. **Weirong Liu:** Writing – review & editing, Visualization. **Chao Hu:** Formal analysis, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was partially supported by the Hunan Provincial Department of Education Scientific Research Project (No. 23B0898)

Data availability

Data will be made available on request.

References

[1] Zhao H, Guo W. Coordinated control method of multiple hybrid energy storage systems based on distributed event-triggered mechanism. Int J Electr Power Energy Syst 2021;127:106637.
[2] Wu Y, Huang Z, Zheng Y, Liu Y, Li H, Che Y, Peng J, Teodorescu R. Spatial–temporal data-driven full driving cycle prediction for optimal energy management of battery/supercapacitor electric vehicles. Energy Convers Manage 2023;277:116619.
[3] Liu S, Miao H, Li J, Yang L. Voltage control and power sharing in DC microgrids based on voltage-shifting and droop slope-adjusting strategy. Electr Power Syst Res 2023;214:108814.
[4] Kumar J, Agarwal A, Agarwal V. A review on overall control of DC microgrids. J Energy Storage 2019;21:113–38.
[5] Dagar A, Gupta P, Niranjan V. Microgrid protection: A comprehensive review. Renew Sustain Energy Rev 2021;149:111401.
[6] Li Y, Zhang Z, Dragičević T, Rodriguez J. A unified distributed cooperative control of DC microgrids using consensus protocol. IEEE Trans Smart Grid 2020;12(3):1880–92.
[7] Zeng Y, Zhang Q, Liu Y, Zhuang X, Guo H. Hierarchical cooperative control strategy for battery storage system in islanded DC microgrid. IEEE Trans Power Syst 2021;37(5):4028–39.
[8] Kirakosyan A, El-Saadany EF, El Moursi MS, Yazdavar AH, Al-Durra A. Communication-free current sharing control strategy for DC microgrids and its application for AC/DC hybrid microgrids. IEEE Trans Power Syst 2019;35(1):140–51.
[9] Lin P, Zhao T, Wang B, Wang Y, Wang P. A semi-consensus strategy toward multi-functional hybrid energy storage system in DC microgrids. IEEE Trans Energy Convers 2019;35(1):336–46.
[10] Chen X, Shi M, Zhou J, Chen Y, Zuo W, Wen J, He H. Distributed cooperative control of multiple hybrid energy storage systems in a DC microgrid using consensus protocol. IEEE Trans Ind Electron 2019;67(3):1968–79.
[11] Lai J, Lu X, Yu X, Yao W, Wen J, Cheng S. Distributed multi-DER cooperative control for master-slave-organized microgrid networks with limited communication bandwidth. IEEE Trans Ind Inf 2018;15(6):3443–56.
[12] Hosseinzadeh M, Salmasi FR. Robust optimal power management system for a hybrid AC/DC micro-grid. IEEE Trans Sustain Energy 2015;6(3):675–87.
[13] Sen S, Kumar V. Microgrid control: A comprehensive survey. Ann Rev Control 2018;45:118–51.
[14] Alavi SA, Mehran K, Vahidinasab V, Catalão JP. Forecast-based consensus control for DC microgrids using distributed long short-term memory deep learning models. IEEE Trans Smart Grid 2021;12(5):3718–30.
[15] Mokhtar M, Marei MI, El-Sattar AA. Improved current sharing techniques for DC microgrids. Electr Power Compon Syst 2018;46(7):757–67.
[16] Zeng Y, Zhang Q, Liu Y, Zhuang X, Lv X, Wang H. An improved distributed secondary control strategy for battery storage system in DC shipboard microgrid. IEEE Trans Ind Appl 2022;58(3):4062–75.
[17] Zhang Q, Zeng Y, Liu Y, Zhuang X, Zhang H, Hu W, Guo H. An improved distributed cooperative control strategy for multiple energy storages parallel in islanded DC microgrid. IEEE J Emerg Sel Top Power Electron 2021;10(1):455–68.
[18] Cao D, Hu W, Zhao J, Huang Q, Chen Z, Blaabjerg F. A multi-agent deep reinforcement learning based voltage regulation using coordinated PV inverters. IEEE Trans Power Syst 2020;35(5):4120–3.
[19] Barbalho P, Lacerda V, Fernandes R, Coury D. Deep reinforcement learning-based secondary control for microgrids in islanded mode. Electr Power Syst Res 2022;212:108315.
[20] Yang Q, Wang G, Sadeghi A, Giannakis GB, Sun J. Two-timescale voltage control in distribution grids using deep reinforcement learning. IEEE Trans Smart Grid 2019;11(3):2313–23.
[21] Afifi MA, Marei MI, Mohamad AM. Reinforcement-learning-based virtual inertia controller for frequency support in Islanded microgrids. Technologies 2024;12(3):39.
[22] Gronauer S, Diepold K. Multi-agent deep reinforcement learning: a survey. Artif Intell Rev 2022;1–49.
[23] Fan Z, Zhang W, Liu W. Multi-agent deep reinforcement learning-based distributed optimal generation control of DC microgrids. IEEE Trans Smart Grid 2023;14(5):3337–51.
[24] Wu Z, Li Y, Gu W, Dong Z, Zhao J, Liu W, Zhang X-P, Liu P, Sun Q. Multi-timescale voltage control for distribution system based on multi-agent deep reinforcement learning. Int J Electr Power Energy Syst 2023;147:108830.
[25] Sun X, Qiu J. Two-stage volt/var control in active distribution networks with multi-agent deep reinforcement learning method. IEEE Trans Smart Grid 2021;12(4):2903–12.
[26] Xia Y, Xu Y, Wang Y, Mondal S, Dasgupta S, Gupta AK. Optimal secondary control of islanded AC microgrids with communication time-delay based on multi-agent deep reinforcement learning. CSEE J Power Energy Syst 2022;9(4):1301–11.
[27] Gu Y, Cheng Y, Chen CP, Wang X. Proximal policy optimization with policy feedback. IEEE Trans Syst Man Cybern: Syst 2021;52(7):4600–10.

[28] Schulman J, Levine S, Abbeel P, Jordan M, Moritz P. Trust region policy optimization. In: International conference on machine learning. PMLR; 2015, p. 1889–97.
[29] Yu C, Velu A, Vinitsky E, Gao J, Wang Y, Bayen A, Wu Y. The surprising effectiveness of ppo in cooperative multi-agent games. Adv Neural Inf Process Syst 2022;35:24611–24.
[30] Cabrane Z, Kim J, Yoo K, Ouassaid M. HESS-based photo-voltaic/batteries/supercapacitors: Energy management strategy and DC bus voltage stabilization. Sol Energy 2021;216:551–63.
[31] Zhang Q, Zeng Y, Hu Y, Liu Y, Zhuang X, Guo H. Droop-free distributed cooperative control framework for multisource parallel in seaport DC microgrid. IEEE Trans Smart Grid 2022;13(6):4231–44.
[32] Chen D, Chen K, Li Z, Chu T, Yao R, Qiu F, Lin K. Powernet: Multi-agent deep reinforcement learning for scalable powergrid control. IEEE Trans Power Syst 2021;37(2):1007–17.